MACHINE LEARNING PROJECT

Polynomial Regression on Manufacturing Dataset

Predicting product quality ratings using process variables and non-linear feature transformation

Project Overview

Core Objective

Model complex non-linear relationships between industrial parameters and final manufacturing output quality.

Methodology

Degree-2 polynomial feature expansion combined with standard linear regression and data scaling.

Target Metric

Accurate prediction of Quality Ratings to streamline automated quality control pipelines.

Dataset Characteristics

3,957 Records
Clean production observations.

Temperature
Primary operating metric.

Pressure
Process chamber pressure.

Fusion Metric
Material fusion tracking.

Transformation
Structural transformation data.

Quality Rating
Target target variable.

Step-by-Step Workflow

01

Setup & EDA

Library loading and correlation checks.

02

Data Split

80-20 train-test partition.

03

Transformation

Degree-2 polynomial features & scaling.

04

Train & Evaluate

Model fit and performance scoring.

Model Performance Results

0.4684

R-squared (R^2) Score

10.42

Root Mean Squared Error (RMSE)

Conclusions & Future Steps

Baseline Established

Degree-2 polynomial features successfully capture basic non-linear trends in manufacturing data.

Hyperparameter Tuning

Experiment with degree-3 polynomials or alternative regularization methods (Ridge/Lasso) to reduce error.

Feature Expansion

Incorporate fusion and transformation metrics directly into feature matrix for enhanced predictive power.

Thank You

Questions & Discussion